

- **Privid: Practical, Privacy-Preserving Video Analytics Queries by Frank Cangialosi et al.**

This paper introduces a privacy-preserving video analytics system based on differential privacy. It protects sensitive information while allowing useful analytics, achieving a balance between privacy and utility in surveillance systems [24].

- **ByteTrack: Multi-Object Tracking by Associating Every Detection Box by Yifu Zhang et al.**

This paper proposes an advanced multi-object tracking algorithm that improves tracking accuracy by considering both high and low confidence detection boxes. It enhances tracking performance in crowded and complex environments, making it highly suitable for surveillance applications [25].

- **AADC-Net: A Multimodal Deep Learning Framework for Automatic Anomaly Detection in Real-Time Surveillance by Duc Tri Phan, Vu Hoang Minh Doan, Jaeyeop Choi, Byeongil Lee, Junghwan Oh**

This paper proposes AADC-Net, a multimodal deep learning framework designed for real-time anomaly detection in surveillance systems. The model integrates vision-language models and object detection techniques to overcome dataset limitations and improve detection accuracy. It achieves strong performance with high AUC and precision on benchmark datasets such as UCF-Crime and XD-Violence. The system is capable of real-time deployment and reduces human effort while improving reliability in anomaly detection tasks [26].

- **A Self-Reasoning Framework for Anomaly Detection Using Video-Level Labels by Muhammad Zaigham Zaheer, Arif Mahmood, Hochul Shin, Seung-Ik Lee**

This paper introduces a weakly supervised anomaly detection framework that uses video-level labels instead of frame-level annotations. The model generates pseudo-labels through clustering of spatio-temporal features, enabling better handling of noisy labels. This approach reduces annotation effort while improving detection accuracy and demonstrates strong performance on real-world datasets like UCF-Crime and ShanghaiTech [27].

- **A Review of Deep Learning-Based Human Activity Recognition on Benchmark Video Datasets by Vijeta Sharma, Manjari Gupta, Anil Kumar Pandey, Deepti Mishra, Ajai Kumar**

This paper presents a comprehensive survey of deep learning techniques used in human activity recognition (HAR). It categorizes approaches into CNN-based and RNN-based methods and evaluates their performance on benchmark datasets. The

study highlights key challenges such as dynamic backgrounds, computational complexity, and the need for large datasets, while also discussing future research directions in HAR for surveillance applications [28].

- **VPN: Learning Video-Pose Embedding for Activities of Daily Living by Srijan Das, Saurav Sharma, Rui Dai, François Brémont, Monique Thonnat**

This paper proposes a Video-Pose Network (VPN) that combines pose information and RGB data for improved activity recognition. By embedding both modalities into a shared space and applying attention mechanisms, the model captures fine-grained spatio-temporal patterns. It improves recognition accuracy for subtle and similar human actions, which is critical in surveillance systems [29].

- **Violence Detection From Industrial Surveillance Videos Using Deep Learning by Hamza Khan, Xiaohong Yuan, Letu Qingge, Kaushik Roy**

This paper introduces a three-stage deep learning framework for detecting violent activities in surveillance videos. It uses a lightweight CNN for person detection, followed by a 3D-CNN for extracting spatio-temporal features, and a classifier for final prediction. The system improves computational efficiency by processing only relevant frames and enables real-time alert generation for safety applications [30].

- **Understanding the Challenges and Opportunities of Pose-Based Anomaly Detection by Ghazal Alinezhad Noghre, Armin Danesh Pazho, Vinit Katariya, Hamed Tabkhi**

This paper analyzes pose-based anomaly detection techniques, emphasizing their advantages in privacy preservation and computational efficiency. It also discusses limitations such as noisy skeleton data and lack of rich features. The study highlights the need for better datasets and improved models for real-world deployment [31].

- **MMNet: A Model-Based Multimodal Network for Human Action Recognition in RGB-D Videos by Bruce X.B. Yu, Yan Liu, Xiang Zhang, Sheng-hua Zhong, Keith C.C. Chan**

This paper proposes a multimodal network that combines RGB and skeleton data for improved action recognition. By leveraging complementary features from multiple modalities, the model enhances accuracy in recognizing complex human activities. It demonstrates superior performance across multiple benchmark datasets [32].

- **Towards a Resource Efficient and Privacy-Preserving Framework for Campus-Wide Video Analytics-Based Applications by Ankur Gupta, Purnendu Prabhat**

This paper presents a framework for efficient and privacy-preserving video analytics in large-scale surveillance systems. It focuses on reducing computational load, storage requirements, and network bandwidth while ensuring privacy protection. The framework is particularly useful for smart campuses and large-scale deployments [33].

- **Weakly-Supervised Video Anomaly Detection with Robust Temporal Feature Magnitude Learning by Tian, Guansong Pang, Yuanhong Chen, Rajvinder Singh, Johan W. Verjans, Gustavo Carneiro**

This paper introduces a weakly supervised anomaly detection approach using temporal feature magnitude learning (RTFM). It improves detection of subtle anomalies by focusing on important video snippets and modeling temporal dependencies. The method outperforms several state-of-the-art approaches on multiple datasets [34].

- **Understanding Ethics, Privacy, and Regulations in Smart Video Surveillance for Public Safety by Babak Rahimi Ardabili, Armin Danesh Pazho, Ghazal Alinezhad Noghre, Christopher Neff, Arun Ravindran, Hamed Tabkhi**

This paper explores ethical and privacy challenges in smart video surveillance systems. It emphasizes the importance of designing systems that balance public safety with individual privacy. The study discusses regulatory frameworks and highlights the need for privacy-aware AI models [35].

- **The Ethics of Facial Recognition Technologies, Surveillance, and Accountability in an Age of AI by Denise Almeida, Konstantin Shmarko, Elizabeth Lomas**

This paper analyzes ethical and legal issues surrounding facial recognition technologies. It compares regulatory frameworks across different regions and highlights concerns related to privacy, accountability, and human rights. The study emphasizes the need for transparency and strict regulations in surveillance systems [36].

- **Self-Supervised Sparse Representation for Video Anomaly Detection by Jhih-Ciang Wu, He-Yen Hsieh, Ding-Jie Chen, Chiou-Shann Fuh, Tyng-Luh Liu**

This paper proposes a self-supervised framework for anomaly detection using sparse representation and dictionary learning. It models normal patterns and identifies anomalies based on reconstruction errors. The method achieves state-of-the-art performance in both unsupervised and weakly supervised settings [37].

- **Deep Learning Based Anomaly Detection in Real-Time Video by Ahmed Elmetwally, Reem Eldeeb, Samir Elmougy**

This paper presents a deep learning-based anomaly detection system using I3D-ResNet50 and Multiple Instance Learning. It processes video snippets to generate anomaly scores and achieves good performance on large datasets. The system improves real-time detection accuracy and efficiency [38].

- **RWF-2000: An Open Large Scale Video Database for Violence Detection by Ming Cheng, Kunjing Cai, Ming Li**

This paper introduces the RWF-2000 dataset, a large-scale dataset containing 2000 real-world surveillance videos for violence detection. It also proposes a deep learning model combining 3D-CNN and optical flow to achieve high detection accuracy. The dataset plays a crucial role in benchmarking violence detection systems [39].

Synthesis of Existing Research

Over the years, video surveillance systems have evolved significantly with the integration of artificial intelligence and deep learning techniques. Earlier works primarily focused on assisting human operators in monitoring environments, such as AI-based systems designed for airport surveillance that improved efficiency through behavior analysis and event detection [1]. This gradually progressed into more advanced intelligent systems capable of performing automated object detection, tracking, and decision-making in real time [2]. At the same time, comprehensive reviews in this domain highlight how the combination of AI, IoT, and edge computing has transformed surveillance systems by enabling faster and more accurate detection [3]. However, these studies also point out that despite technological improvements, practical challenges such as scalability, data dependency, and privacy still remain unresolved.

As research moved forward, a large number of studies began focusing on deep learning-based anomaly detection techniques. Models based on convolutional neural networks, generative adversarial networks, and transformer architectures have shown strong performance in identifying unusual patterns in video data [4], [11]. These methods are highly effective in controlled environments and benchmark datasets, but in real-world applications, they often require large computational resources and high-quality labeled data. This makes them difficult to deploy in real-time scenarios, especially in systems with limited processing power. Similarly, approaches that use visual attention mechanisms or skeleton-based action recognition improve accuracy by focusing on important regions or human movements [12], [9], yet they still face issues when applied to diverse and unpredictable environments.

To overcome the limitations of centralized processing, many researchers have explored edge computing and distributed architectures. Systems such as AIoT-based surveillance frameworks and multi-camera edge analytics platforms aim to reduce latency and bandwidth consumption by processing data closer to the source [5], [6]. In addition,

edge-cloud collaborative models further enhance scalability and enable efficient real-time detection across multiple locations [13]. While these approaches significantly improve responsiveness, they often introduce additional complexity in system design and may not always be feasible for large-scale or cost-sensitive deployments.

Another important direction in recent research is reducing the dependency on labeled datasets. Techniques such as weakly supervised learning and temporal feature-based anomaly detection attempt to learn patterns using minimal annotations, making them more practical for real-world applications [27], [34]. Multimodal approaches like AADC-Net further enhance detection capabilities by combining multiple data representations and improving robustness [26]. Even though these methods show promising results, accurately detecting rare or complex anomalies in real-time environments still remains a difficult problem due to the unpredictable nature of abnormal events.

Human activity recognition and pose-based methods have also become key components of intelligent surveillance systems. Techniques using graph convolutional networks, pose embeddings, and multimodal learning help in understanding human behavior more effectively [9], [29], [32]. Research surveys in this area highlight challenges such as occlusion, dynamic backgrounds, and computational overhead [28]. While these approaches improve behavioral analysis, they are not always reliable in real-world conditions where multiple variables affect performance.

In parallel, privacy and security have emerged as critical concerns in surveillance applications. Several works focus on privacy-preserving video analytics, encryption frameworks, and differential privacy techniques to protect sensitive information [20], [22], [24]. Ethical studies further emphasize the importance of responsible AI usage and regulatory compliance in surveillance systems [35], [36]. However, implementing strong privacy measures often comes at the cost of reduced accuracy or increased system complexity, making it difficult to achieve an ideal balance.

Additionally, application-specific research such as violence detection, traffic monitoring, and dataset development demonstrates the practical importance of surveillance systems in improving public safety [30], [7], [39]. While these solutions perform well in specific scenarios, they are often limited in adaptability and cannot be easily generalized across different environments or use cases.

Taking all these studies into account, it becomes clear that although significant progress has been made, existing systems still face several important limitations. Many models are computationally intensive and not suitable for real-time edge deployment, while others lack scalability or depend heavily on large labeled datasets [4], [11], [27]. At the same time, most approaches address individual problems in isolation rather than providing a unified and efficient surveillance solution. Privacy-preserving methods, although important, are still not seamlessly integrated with high-performance detection systems [20], [35]. Moreover, real-world challenges such as lighting variations, occlusion, and dynamic environments continue to affect system reliability and accuracy [28].

Therefore, there is a clear need for a more balanced and practical approach that combines efficiency, scalability, and privacy while maintaining strong detection performance. The proposed system aims to address these gaps by developing an intelligent surveillance framework that supports real-time anomaly detection, reduces computational overhead, and adapts effectively to real-world conditions, making it more suitable for practical deployment.

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